Automation in Construction 85 (2018) 107–123

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Review

Automated point mapping for building control systems: Recent advances and future research needs

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ARTICLE INFO

Keywords:
Building automation systems
Building control
Machine learning
Point mapping
Metadata
Semantics

ABSTRACT

This paper presents a review of recent research and development on methodologies relevant to automating mapping of points in building control systems and between building control systems and external or replacement software and hardware. Manual point mapping is labor intensive and costly, presenting a major impediment to innovations in building control (e.g., automated fault detection and diagnostics, self-healing, and automated commissioning for existing building control systems). The methods reviewed focus on classifying building control system points, especially sensor classifications by sensor type. Fewer publications address other important aspects of the point mapping problem, such as discovering spatial and functional relationships among points, relationships between control system points, physical systems, and equipment, and between various equipment and the systems of which they are part, and discovering metadata, normalizing it to a common namespace, and assigning the metadata to control system points. To motivate further development of new automated point mapping approaches, we identify many research questions organized into four key technical needs: 1) a complete solution and underlying problem formulation, 2) alignment of methods with the actual point mapping problem, 3) test cases, data sets for testing, explicit test procedures, and consistent performance metrics for reporting testing and evaluation results, and 4) understanding of the applicable data space to ensure future adaptability of automated BAS point mapping.

1. Introduction

Building automation systems (BASs) serve buildings that represent about 42% of the total floor area in the U.S. commercial buildings sector [1,2]. Seventy-nine percent of this floor area is in buildings with areas > 4645 m^2 (50,000 ft^2) [1]. Furthermore, the fraction of floor area served by BASs significantly decreases as the size of buildings decreases; however, 75% of the number of U.S. buildings with BASs are buildings with floor areas of 4645 m^2 and less, and the average size of buildings with BASs is 4407 m^2 (47,440 ft^2) [1,2]. Still, only 11% of all commercial buildings with floor areas ≤ 4645 m^2 have BASs compared to 70.3% of all buildings with floor areas > 18,580 m^2 (200,000 ft^2) [1,2]. To address the absence of BAS technology in commercial buildings smaller than 4645 m^2, work is currently underway to extend BAS functionality cost-effectively to these smaller commercial buildings, which generally do not have BASs presently (see, for example, [3,4]).

A significant effort has been underway for > 3 years by the U.S. Department of Energy and others to bring BAS technology at significantly lower cost than previously possible to such smaller buildings. This effort is centered around the flexible, low-cost, modular, open-source controls platform, VOLTTRON™ [5,6], for which several building control and analytic applications have already been built (c.f., [4,7], and http://www.bemoss.org/).

Extension of BAS technology to small commercial buildings will be necessary, if as many researchers, policy analysts, and utility personnel believe, buildings will serve as active participants in managing the emerging modernized electric power grid by controlling loads in response to conditions on the grid or incentive signals sent by the grid, such as prices [8–13]. Success of these developments should lead to the emergence of many third-party applications for commercial buildings of all sizes that rely on building control systems for data.

BASs consist of hardware and software integrated into a single architecture and are intended to keep building occupants comfortable, control systems and equipment (primarily for heating, ventilating and air-conditioning), monitor and track the performance of various systems, and provide notifications to building operators of device and

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http://dx.doi.org/10.1016/j.autcon.2017.09.013
Received 3 January 2017; Received in revised form 4 July 2017; Accepted 13 September 2017
Available online 19 October 2017

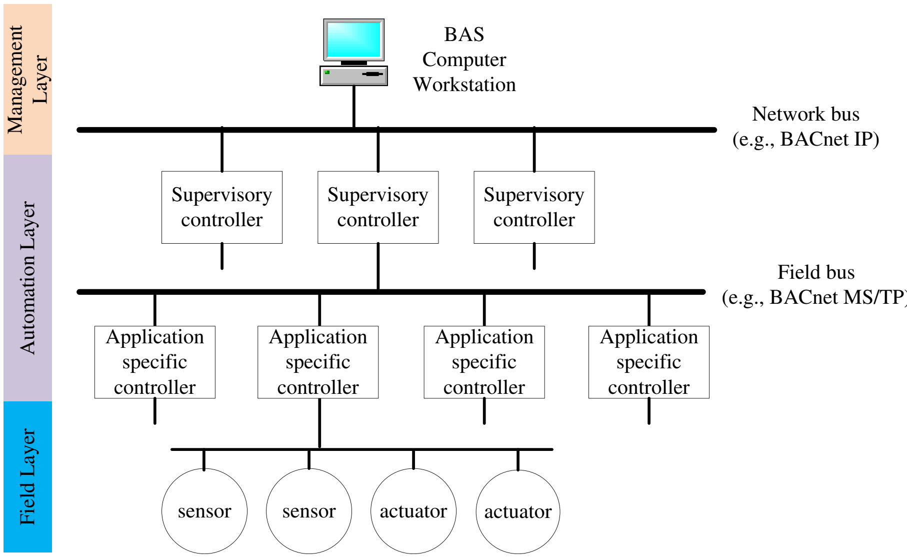

Fig. 1. Building automation system architectural structure.

system faults and failures through alarms (e.g., alarms indicating that a zone air temperature is outside its normal range and that air filters are clogged with dirt).

The classic BAS architectural structure defines three layers (Fig. 1), which from bottom to top are the field layer, automation layer, and management layer [14]. The field layer consists of various sensors and actuators that interface to physical processes. The sensors (e.g., thermistors and flow meters) measure the condition or status of physical processes and devices. The measured data are gathered at direct digital control stations, which are usually application-specific controllers in the automation level. These controllers process the sensed data and command the actuators to take appropriate actions (e.g., regulating the position of a valve to control a water flow rate). The automation layer may also include supervisory controllers, which generally perform global control functions such as scheduling, alarming, and recording data. Field buses are used to communicate between application-specific controllers and supervisory controllers, while networks based on the Internet Protocol (IP) are used for communication between supervisory controllers. The management layer has computer stations or portable devices used to manage and monitor the entire system. Through a unified user interface, building operators can modify schedules and control loop parameters, archive data for long-term historical analysis, and generate reports. In the context of building automation systems, sensor measurements, actuator outputs, and adjustable parameters in control algorithms are often called BAS points. Fig. 1 shows a classic BAS architecture normally seen in the field, but other configurations exist, including some with sensors connecting directly to field buses or application specific controllers using IP.

Correctly installed and well-operated BASs can bring many benefits to building owners and operators, such as increasing occupant comfort, reducing occupant complaints, lowering utility costs, decreasing maintenance costs, and enhancing property values [15–17]. However, control systems are currently largely installed by an error-prone manual process, during which the associations among sensors, actuators, and their connected controllers, between application-specific controllers and their supervisory controllers, and between different peer controllers must be correctly established. High costs for installation, programming, and commissioning of BASs ($1.0 to $6.0 per square foot [17]), especially for specialized labor, have historically been one of the major reasons why many commercial buildings – and in particular the many medium and small commercial buildings – do not have BASs. Automating the point association (i.e., mapping) process will help ensure error-free, less labor intensive, consistent, and complete implementation of control algorithms during installation and thereby significantly lower the total installation costs.

BASs can provide a platform on which new energy-saving technologies can be deployed, first as external software connected to them and later as capabilities integrated into BASs. Software applications using data from BASs have shown substantial benefits in enhancing building operational efficiency, occupant comfort, and grid services [18,19]. These software applications provide functionalities, such as remote monitoring of building and equipment operation, energy information management, fault detection and diagnostics, adaptive control, self-correcting control, and other innovative functions. These applications usually connect to and rely on BASs as their source of data and a means of implementing control actions so as to avoid the expense of a separate network. Connecting to BASs though leads to the need to *map* BAS points to points in these external software applications, which is ordinarily done manually or using a customized semi-automated process.

Points in a BAS ordinarily have little, if any, computer readable metadata that identify point functions, characteristics, and relationships to other points in the control system, devices in the physical (e.g., heating, ventilation and air-conditioning [HVAC]) systems, and spaces in the building. When metadata are available, they are often embedded in point names or in extended text descriptions in fields associated with a point. This suggests that metadata could be automatically retrieved from BAS point names or other text associated with them by simple parsing of text strings. However, because no naming convention is yet widely used, point names and extended descriptions are inconsistent across buildings and BASs (even those on a specific campus). Point naming is often ad hoc with names differing in the information included, terminology used, and abbreviating of terms. Even though some work towards the use of standardized point tags is ongoing (e.g., [20]), current tagging schemes cannot represent all information required by third-party software applications [21]. Tags can be embedded in point names or captured in a table, for example, associated with each point. The present chaotic naming makes the process of parsing text to determine metadata difficult to automate and often requiring involvement of an expensive expert (e.g., control system implementer or building engineer). The resulting high cost of implementing software that relies on mapping of BAS points often makes the software applications prohibitively expensive and unable to penetrate the marketplace. For example, it took 3 days for a building expert to manually map 9820 BAS points from 6 buildings to an enterprise energy management system [22].

Successful automation of the BAS point mapping process would overcome a key barrier to the cost-effective use of third-party devices and software that require data communication to and from existing building control systems. It would also enable a path to practical implementation for many new building operation capabilities that would save energy, reduce greenhouse gas emissions, and deliver more reliable services to building occupants and owners. Automation of the process of connecting BAS data points to third-party devices and software could potentially minimize their installation cost, making them affordable to the small and medium commercial building subsectors. One way that the automated point mapping capability could be made available is as open source software, so that application developers can embed it for a very low cost (just the cost of making it part of the application). That approach could lead to the cost of value-added applications to BASs for commercial buildings decreasing precipitously in future years, thus increasing the affordability of third-party applications that improve system performance, save energy, and decrease building operating costs, overcoming cost barriers for small- and medium-size buildings especially.

Recognizing the importance of point mapping for BAS setup and third-party software integration, researchers have investigated potential approaches over the last several years to solve elements of the automated point mapping problem. This paper provides a review of recent advances, identifies research and development needs to realize scalable automated point mapping, and formulates research questions intended to spur development of new effective automated point mapping approaches.

2. Point mapping use cases

Use cases provide examples of how automated point mapping could be used. As such, they serve to identify potential applications and requirements for point mapping. Three representative examples spanning a range of applications follow. The procedures in each use case are based on the authors' review of relevant literature and discussions with BAS installers, but they by no means represent the only sequences of steps that could be used.

2.1. New BAS set up and configuration

A representative work flow for setting up and configuring a BAS consists of the following steps [23,24]:

1) Set up the communication network. This step involves connecting operator workstations, network controllers, and other networking devices via cables or wirelessly to form the layers of the BAS network (e.g., the management network and the automation network layers).
2) Install and wire the application-specific controllers. This generally involves installing and connecting application-specific controllers (APCs) to the automation layer field bus network (see Fig. 1) and wiring sensors and actuators to their APCs.
3) Select, install, and configure programs and test the application-specific controllers. In this step, control programs are downloaded to each device and configured by setting network parameters, specifying point assignments, and setting parameter values for control algorithms. The controllers are tested to ensure proper operation. These tasks are usually performed by the controller installer using configuration software tools, one controller at a time as it is installed.
4) Install and wire the supervisory controllers. This step includes installing building-level supervisory controllers and wiring them to their underlying application-specific controllers and the computer workstation for the BAS.
5) Configure, program and test the supervisory controllers. This process involves creating schedules, data trending, reports, and event handling. In addition, it includes configuring and programming the supervisory controllers to implement control functions.
6) Commission the entire system. Commissioning includes tuning the control parameters and testing the entire system under various conditions to verify that it works as intended.

Point mapping is needed in Step 2 to ensure the sensors and actuators are wired correctly to each application-specific controller. Point mapping is also needed in Step 5 when programming supervisory controllers and binding data points to BAS graphics. Because the supervisory controllers need to be programmed specifically for individual applications, the data inputs from application-specific controllers or other peer supervisory controllers must be defined and mapped correctly. In a variation of this use case, in the longer-term future, the hardware system could be completely installed without manual programming and assignment of points to individual devices. Instead the entire process would be automated so that identifying, naming, and attaching metadata to points would be performed automatically, leading to a much less labor intensive and costly process for new BAS installations.

2.2. Adding or replacing controllers in an existing BAS

A controller needs to be replaced when it fails or no longer has the desired functions and features. New controllers also need to be added to an existing BAS when a building is expanded or renovated and additional HVAC equipment is installed to accommodate changes in spatial arrangement and functionality. In both situations, the added controller may have new points or points named differently than those that were in the replaced controller. These points need to be mapped to existing BAS points. The work flow for adding or replacing a controller generally consists of the following steps:

1) Discover BAS points in the existing system.
2) Create new BAS points required by the additional or replacement controller.
3) Define the semantic metadata (e.g., that a specific temperature sensor measures the temperature of chilled water entering the coil of a specific air handler) for all BAS points, including both existing ones from Step 1 and newly added ones from Step 2.
4) Using the metadata, establish the relationships between the points in the newly added controller and those in the existing BAS.
5) Test and verify that the newly added controller and the existing components that rely on data from points of the newly added controller work correctly.

Automated point mapping could use automated point discovery (in Step 1), which is a functionality provided in many of today's BASs that use standardized communication protocols such as BACnet [25]. More importantly, automated point mapping is needed in Steps 3 and 4 to efficiently distinguish among sensors, actuators, and control points and assign equalities between each point in the newly added controller and its corresponding point in the existing BAS.

2.3. Connecting third-party add-on software to an existing BAS

Third-party software applications are usually (initially) separate from BASs but require connections to BASs for the purpose of gathering data and transmitting control signals back to building systems, which leads to the need to map BAS points to points in the external software. The work flow of connecting third-party software to a BAS consists of the following steps:

1) Discover and retrieve all points in the BAS.
2) Filter out the discovered points that are obviously irrelevant to the third-party software application. For example, if the third-party software needs only analog inputs, all points indicated as analog outputs, binary inputs, and binary outputs are regarded as obviously irrelevant and can be filtered out.

3) Assign semantics to each BAS point left from Step 2.
4) Establish relationships between the inputs and outputs of the third-party software and the corresponding BAS points.
5) Run the third-party software and verify that it works correctly.

Point mapping needs are similar to those defined in the previous use case (Section 2.2). As part of point mapping, point discovery is needed in Step 1, semantic metadata needs to be defined in Step 3, and equality relationships need to be established for connecting the third-party software and the BAS in Step 4. Automation of the point mapping function would simplify the process of connecting a BAS to third-party software to provide data needed by the application, considerably reducing or eliminating the need for an expert for point mapping, and decreasing the cost of new software installation.

3. Technical challenges and objectives for automated point mapping

Every BAS point has at least one name, which is a human interpretable text string. In some systems, a cryptic name is used internally and another name is also used to make points (somewhat) more easily understood and interpreted by humans. Meaningful point names provide useful information for building operators, control system design engineers, and control system installers for use in their job functions, such as maintaining, modifying, and integrating components into building control systems. In reality, however, several challenges exist to realize automated point mapping based merely on point names. First, point names and other point attributes (e.g., units and object types) were not created with the intention of being machine decodable. Secondly, because no widely-used standardized naming conventions exist (or, if they did, would unlikely be used widely at any time soon), point names can have multiple and inconsistent uses of schemas, abbreviations, and delimiters across different buildings and even among points in the same BAS and building. Thirdly, the information captured in point names and other point attributes is often incomplete and incorrect as a result of the judgement exercised or errors made during configuration.

Effective automated mapping must be a scalable, cost-effective, and high-performance solution to map points to one another and to the data requirements of third-party software and hardware. Scalable in the context of this paper is defined as capable of being readily applied to a large segment of the entire population of commercial buildings; i.e., the solution is generalizable to different building types, BASs, and third-party applications. Cost-effective means that the cost of the automated mapping solution is recovered in a period of time that satisfies the rate of return expected by the BAS installer or the building owner, where the cost includes both the capital cost of the solution as well as the installation and commissioning costs. Software delivering the solution needs to be deployed easily without much customization and user intervention. High-performance signifies that the automated mapping solution provides accurate semantics for most BAS points while mapping a high percentage (e.g., > 95%) of all points for which mappings are needed.

Identification of ideal characteristics for an automated point mapping solution requires application of logic and judgement to information on the process being automated, the sources of value to be derived from automation, and the factors most affecting the performance and cost of the process, among others. As a result, the ideal characteristics, which follow, implicitly depend on the authors' judgements, are inherently uncertainty, and should be interpreted accordingly. Yet, they can serve as targets for the ultimate performance sought in research on this topic.

- *Agnostic to BASs*. To be widely applicable, automated point mapping must be agnostic to the specific BAS, i.e., it must apply across BAS vendors, BAS models or versions, vintage, point naming practices, etc. As mentioned previously in Section 2.2, automated discovery is part of automated mapping. The purpose of automated discovery is to find all BAS points and their attributes (e.g., names) in existing BASs. If the automated mapping functionality is implemented in external software, communication must first be established between the BAS and the auto-mapping software, regardless of the BAS vendor and the communication protocol used.
- *Agnostic to third-party application software*. The point information required by third-party software varies significantly. To accommodate the diversified needs for point information, automated mapping must be accompanied by a mechanism to represent point semantics (i.e., metadata). Automated mapping will need to store the metadata associated with every point; therefore, it must connect to and store point names and associated metadata in an open building data model that all users can freely access. The open data model, including a metadata representation scheme [26] can be regarded as the unified interface to third-party software. In the future, an open building data model should be standardized.
- *Support all point semantic information needed by third-party software, hardware, and the control system itself*. Bhattacharya et al. [21,27] report a variety of metadata required by third-party software. Such information includes point types, the associations between equipment and points, the functional relationships between equipment, and the spatial relationships among points, equipment, and building spaces. All this information should be supported by automated point mapping. An effective automated mapping method needs to generate a clear system topology for building HVAC and other systems. An example system structure for a built-up variable-air-volume system is shown in Fig. 2, which illustrates 1) the connections between a chiller and the air handling units (AHUs) it serves; 2) the connections between each AHU and the served terminal boxes (TBs); 3) the connections between TBs and spaces; 4) the points belonging

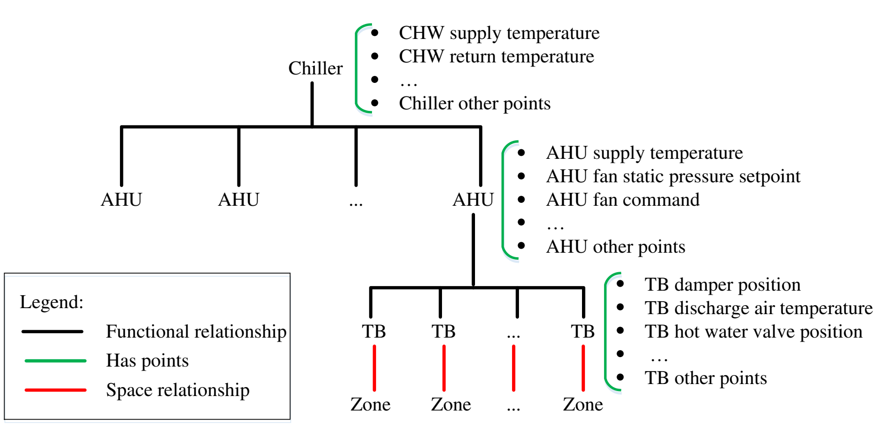

Fig. 2. Example showing the metadata for each point that should result from application of automated mapping to the control system for a commercial VAV system. AHU, CHW, TB, and zone represent air handling unit, chilled water temperature, terminal box, and thermal zones for heating and air-conditioning, respectively.

Table 1
Example BAS point information for a VAV terminal box from the BAS of an operating building. The first column provides an independent description of each point for comparison with the BAS information given in the other columns.

Point description	BAS point information			
	Point name	Units	BACnet object type	Property
Calculated value of actual outdoor-air volumetric flow rate to a specific (current) zone in ft^3/min	Current zone OSA CFM	cubicFeetPerMinute	analogValue	presentValue
Required outdoor-air volumetric flow rate to a specific (current) zone in ft^3/min	Current zone OSA CFM needed	cubicFeetPerMinute	analogValue	presentValue
Command for a specific VAV-box damper position as a percentage of the fully open position command	Damper position	percent	analogValue	presentValue
Command for a specific VAV-box reheat valve position as a percentage of the fully-open valve position command	Htg valve position %	percent	analogValue	presentValue
Measured VAV box airflow rate in ft^3/min	Airflow sensor	cubicFeetPerMinute	analogValue	presentValue
Measured air temperature in °F at location of the thermostat in a specific zone (served by the VAV box)	Room temp	degreesFahrenheit	analogInput	presentValue
Set point for air temperature in °F for the room with the thermostat for the specific zone where the VAV box thermostat is located	Room temp set point	degreesFahrenheit	analogValue	presentValue
Temperature of the air in °F leaving the VAV box	Supply air temperature	degreesFahrenheit	analogInput	presentValue
Sensed occupancy status (occupied or unoccupied) in the current zone	Zone occupancy sensor status	enum	binaryValue	presentValue

to each physical device; 5) other high-granularity point-type information.

- *Accuracy in determination of point semantics near 100%*. Only covered fields (see next bullet on coverage) are included in determination of the accuracy of the model (or method). Incorrectly determined fields (e.g., resulting from assigning an incorrect semantic to a point) can lead to improper control and/or analysis of system performance. For a specific point, the accuracy can be defined as the fraction of the total number of semantic fields for which the method provides the correct metadata (semantic information). The accuracy required for effective automated point mapping depends on the application and, if generalized, should be determined by the community of users, although it likely exceeds 90% or 95% to minimize the number of incorrectly characterized points.
- *Coverage of a large fraction of BAS points*. To address the primary issue of high labor requirements and costs associated with manual point mapping, the automated process must cover a significant fraction of all control system points. Therefore, effective automated mapping must be capable of handling points, irrespective of the signal type (e.g., analog, binary, or multi-state) and signal direction (e.g., sensor outputs, actuator inputs and outputs, and software values). Full coverage is unlikely for points having semantics that appear rarely in building control systems. Points and metadata not covered by automated mapping could be covered by a supplemental manual or semi-automated mapping method, which together with the automated mapping process could cover 100% of all points and metadata.
- *Minimum human involvement*. Once the communication with a BAS is established, the automated mapping solution must provide results with minimal user or expert involvement to minimize the cost of metadata and point mapping.
- *Scalability to most of the commercial buildings sector*. For an automated BAS point mapping method to sufficiently reduce the cost associated with the mapping process, it must generalize to very large numbers of commercial buildings beyond the set used in developing the algorithm so that each point mapping algorithm applies to a significant fraction (e.g., > 10%) of the commercial building population. This requirement would enable a reasonable number of specific mapping algorithms (e.g., < 10) to cover essentially all commercial buildings. Methods requiring training should transfer without significant customization from one building to others; however, no single instantiation (trained version) of a method will likely apply to all commercial buildings, because the commercial buildings sector comprises many building types [e.g., 13 building types based on primary activity in the U.S. [2]]. However, a successful algorithm will likely need to apply to tens to hundreds of thousands of buildings without retraining.

4. Recent advances in automated point mapping

Recognizing the value and need for automatically mapping BAS points to external software applications, several research groups have explored different approaches, including point type classification, metadata transformation, sensor identification, discovery of spatial relationships between sensors and building spaces, and discovery of functional relationships among different equipment.

4.1. Point-type classification

Point-type classification, which attempts to associate each control system point (or in some cases just sensor points) to a specific type (e.g., temperature sensor, fan speed, and damper position), is one of the most studied topics in the field of automated BAS point mapping. Studies formulate and address the problem of BAS point classification differently. Major differences across studies are:

- *The definition of point types*. The appropriate BAS point taxonomy depends on the problem to be solved. However, the level of detail in BAS point taxonomy has a significant impact on the difficulty of classification and the resolution of the results. Detailed point taxonomies usually pose greater challenge to the classifier than high-level, lower resolution, taxonomies. For example, classifying a BAS point as either a temperature sensor or a pressure sensor is generally easier than classifying a sensor as an AHU's return-air temperature sensor or a zone-air temperature sensor.
- *The kind of metadata used for point type classification*. Metadata used in point type classification studies include: 1) text information for points (static data) such as point names or the combination of point names and other point descriptors (e.g., units); 2) point time-series data (dynamic data); and 3) the combined use of text and time-series data. As an example, Table 1 shows several BAS points and their textual information in the BAS for a VAV terminal box in a real building. Based on the BAS points from three buildings, Hong et al. [28] demonstrated that point textual data are good indicators of point types, but they are not consistent across buildings, while time-series data are more consistent across buildings but are poor indicators of point types.
- *The methods and algorithms used for pre-processing, training, and*

classification. Methods vary significantly and are not generally compared across studies. Some methods use a human in the loop, while others do not; some studies report on the process for selecting training data (e.g., random versus hand-picked); results are reported differently (e.g., different metrics) with some authors giving more statistical information than others.

Recent studies of point type classification organized by the kind of metadata used are described in Sections 4.1.1 through 4.1.4, which follow.

4.1.1. Point type classification based on point textual data

Schumann et al. [29] applied linguistic similarity to establish the mapping relationship between BAS point names and the labels used in an enterprise management system (EMS), where the EMS labels can be regarded as equivalent to point types. To calculate the linguistic similarity between the BAS point names and the EMS labels, a marker set is defined to represent the EMS labels that normally consist of acronyms or abbreviations. A dictionary is also defined, which provides the mapping between the marker set and the words in BAS point names as delimited by a special character like the underscore. Using linguistic similarity can reduce the number of EMS labels for consideration before seeking an accurate match. However, in many cases, choosing the EMS label and point name pair with the highest similarity rating does not ensure a correct match. A test on 2099 BAS points from a laboratory building using a dictionary with 629 terms (504 acronyms and 125 markers) from other data sets showed the marker set with the highest similarity corresponding to the correct labeling in only 16% of the cases. This result indicates that use of linguistic similarity will not support fully automated labeling; however, prioritization of matches based on linguistic similarity scores can be used to considerably reduce the number of matches that a user would need to consider manually (a mean of only 7.5% of all possible EMS labels in the evaluation of Schumann et al.).

As noted by Hong et al. [28], the performance of linguistic similarity depends significantly on the coverage and diversity of entries in the dictionary. The dictionary size may become unwieldy when many variations in point naming exist, which is actually the case in the buildings field. In addition, using delimiters to identify words is not reliable because delimiters are often not used at all in BAS point names or are not represented consistently by non-alphanumeric symbols.

Park [30] used a topic model based on Latent Semantic Indexing (LSI) to discover the abstract "topics" that occur in a collection of BAS points, where the topics are represented as discrete probability distributions over predefined words (i.e., a vocabulary). The abstract topics from LSI are processed by a Naïve Bayes classifier to determine the point type. Park used about 19,500 points from three buildings to train a model and tested the model on data from another two buildings having about 6700 points. More than 90% classification accuracy was reported for the points in the two test buildings. The BASs in all five buildings were apparently from the same vendor, which may have increased the similarity in data schemes across the buildings. The classifier performance for BASs from other vendors was not tested.

Hong et al. [28] applied a clustering-based active learning approach for the classification of BAS points into 21 point types (Table 2). Their method preprocesses the original point names first by converting all letters to lowercase and eliminating numerical and special characters. Then, the k-mers[1] for each point name are calculated and used to represent the point feature. Using k-mers as the inputs, BAS points are clustered and classified. To minimize the number of training data, BAS points used for training are selected stepwise following an active learning approach. In each step, an instance is selected for manual labeling of point types based on two criteria: 1) the instance should come from the most informative cluster and 2) the instance should be representative of that cluster. A label propagation mechanism is used to speed up the learning process. The clustering-based active learning approach was tested on three buildings having BASs from different vendors. The results indicated that one building required about 18% of the BAS points to be manually labeled to achieve 90% classification accuracy, while the other two buildings required < 3% of BAS points to be manually labeled with the accuracy being 99%. Comparison of this method with application of four other active learning approaches (i.e., random selection, least margin, pre-clustering, and hierarchical clustering) to the same data shows that the active learning approach of Hong et al. [28] based on Gaussian mixture model clustering and label propagation requires fewer manually labeled points.

[1] A k-mer is all the possible substrings of length k that are contained in a specific string (e.g., a point name).

Table 2
Point categories used by Hong et al. [28,31,32].

Point type category for classification	Hong et al. [28]	Hong et al. [31]	Hong et al. [32]
CO_2 concentration	✓	✓	✓
Humidity	✓	✓	✓
Air pressure	✓		✓
Room temperature	✓	✓	✓
Facility operation status	✓		✓
Facility control	✓		✓
Set point	✓	✓	✓
Air flow volume (i.e., volumetric air flow rate)	✓	✓	✓
Damper position	✓		✓
Fan speed	✓		✓
Hot water supply temperature	✓		✓
Hot water return temperature	✓		✓
Chilled water supply temperature	✓		✓
Chilled water return temperature	✓		✓
Supply air temperature	✓		✓
Return air temperature	✓		✓
Mixed air temperature	✓		✓
Ice tank entering temperature (for ice thermal energy storage)	✓		✓
Ice tank leaving temperature (for ice thermal energy storage)	✓		✓
Occupancy	✓		✓
Timer	✓		✓
Other temperature		✓	

4.1.2. Point type classification based on point time-series data

Studies by Hong et al. [31], Gao et al. [33], and Balaji et al. [34] use time-series data for BAS point type classification. The major assumption that underlies this method is that the patterns in sensor time series from different sensor types are distinguishable using certain statistical features (e.g., percentiles and variances). In Hong et al. [31], the statistical features are extracted in two steps. The time-series data for each BAS point are first segmented into N non-overlapping 45-minute long windows. The median and variance of point readings are calculated for each time window, which leads to a vector of medians and a vector of variances (with vector size N). In the second step, a statistical summary is created, consisting of the minimum, maximum, median, and variance for each of the two vectors from the first step. The resulting feature vector consists of eight variables for each BAS point. The BAS point features are then used as classifier inputs. Hong et al. used the random forest classifier, an ensemble learning method, for the classification of BAS sensor points into six types: room CO_2 concentration, room humidity, set point, VAV air flow rate, room temperature, and temperatures other than room temperature. Using 1936 sensor points from two buildings (with different locations and BAS vendors) each over a one-week period, they performed both intra-building and inter-building tests with different sizes for the training data sets. For the intra-building test, both training and test data come from the same building, while for the inter-building test, training data and test data are from different buildings. The test results show that 1) the classifier performance varies across point types [for example, for one building with 20% of the data used as the training set, the classification accuracy ranges from 50% for the air flow volume (i.e., flow rate) point type to almost 100% for the set point]; 2) for most point types, intra-building tests show higher classification accuracy than inter-building tests (for example, about 20% difference in classification accuracy was observed for the of room CO_2 concentration point type); 3) in general the classifier performance improves with increasing size of the training data set, but the marginal improvement decreases significantly when the size of the training data set increases beyond 20% of the full data set; and 4) the classifier performs much better with the proposed data feature scheme compared to a simplified feature scheme using the median and variance across the entire time series.

Gao et al. [33] performed a study similar to the one by Hong et al. [31], using statistical features extracted from the BAS time-series data for point type classification. For each data point, the feature set includes the mean, median, mode, quartiles, and deciles across the entire time series. Point types are represented with standardized Haystack tags [20]. Eight classification algorithms were investigated, random forests, k-nearest neighbors (kNN), decision trees, Gaussian naive-Bayes, non-linear support vector machines (SVM), logistic regression, AdaBoost (adaptive boosting), and linear discriminant analysis. The classification was performed in two different manners, using composite labels directly as the classifier outputs and using individual tags as an intermediate step in developing composite labels. For example, BAS points indicating “airflow request” can be represented as a composite label “VAV air flow cmd” (where cmd represents command) or the combination of the individual tags “VAV,” “air,” “flow,” and “cmd.” For the case of using individual tags, the final composite label of a BAS point is determined as the one having the smallest Euclidean distance to the vector of predicted probabilities for individual tags. Gao and colleagues tested their approach on historical time-series data from two buildings (one in the U.S. for which data for two months, December and June, with a sampling interval of about 15 min were used and the other in Ireland for which three years of data with varying sampling frequency were used). After some pre-processing to eliminate failed sensors (i.e., sensors for which 95% of the measured values do not change over the entire time series) and sensors with small numbers of samples (i.e., < 1000 over the data collection period), the number of BAS points for classification was decreased to 2354 for the U.S. building and 1678 for the building in Ireland. The analysis for both buildings used 20% of the BAS points as the training set with the remaining points used for testing. Using the F1-score[2] as the performance metric, the test results show that 1) random forests yielded the best performance with composite labels, while kNN performed best for use of individual tags; 2) the random forest classifier achieved a mean F1-score for training data set sizes from 20% to 90% of each of the two building data sets of 0.6 for the U.S. building and 0.75 for the Irish building for point type classification with composite labels; 3) classification performance was insensitive to whether individual tags or composite tags were used; and 4) the classification performance was not acceptable for both training on one building and testing on another building and, for one building, training on data for one season and testing on data for another as indicated by F1-scores close to zero for many point types.

[2] F1 is the harmonic mean of precision and recall, i.e., $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$, which is a balanced value of $F = (1 + \beta^2) \times ((\text{Precision} \times \text{Recall})/(\beta^2 \times \text{Precision} + \text{Recall}))$, where $\beta = 1$ so that Precision and Recall are equally weighted. The F-score can also be interpreted in terms of Type I and Type II errors: $F = (1 + \beta^2) \times (\text{True Positive Rate}) / [(1 + \beta^2) \times (\text{True Positive Rate}) + \beta^2 \times (\text{False Negative Rate}) + \text{False Positive Rate}]$.

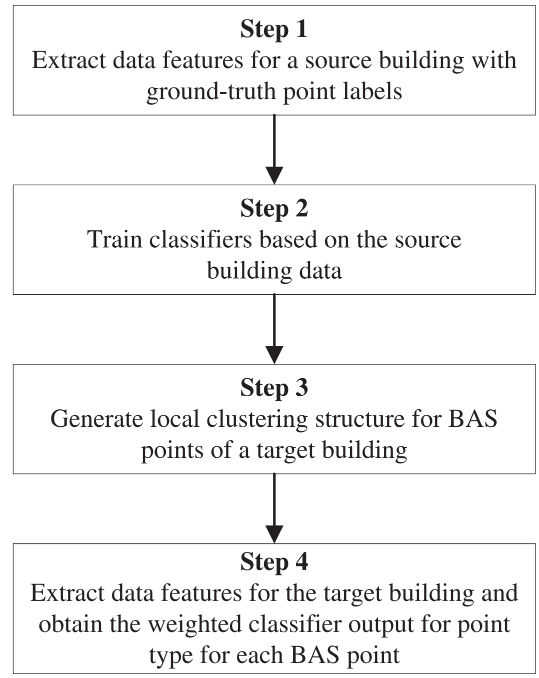

Fig. 3. Primary sub-processes in the approach of Hong et al. [32].

4.1.3. Point type classification based on the combination of point textual data and time-series data

Extending their previous work [28,31], Hong et al. [32] combined the use of textual data and time-series data to classify BAS point types. The primary subprocesses in their approach are shown in Fig. 3.

1) Starting from a source building with the ground-truth labeling of all BAS points, statistical features are extracted from the time-series data. With respect to data feature extraction, two changes were made relative to the earlier study by Hong et al. [31]: i) the time-series data are segmented into one-hour long windows with 50% overlap between consecutive windows and ii) the feature vector includes 44 variables representing the summary (i.e., minimum, maximum, median, and variance) of 11 statistical metrics from the series of one-hour data windows. These 11 metrics are used to cover the following statistical features: extrema (minimum and maximum), average (median and root mean square), quartiles (1st and 3rd quartiles, and inter-quartile range), moments (variance, skewness, and kurtosis), and shape (simple linear regression slope).
2) The data features together with the known BAS point labels are used to train an ensemble of three sensor type classifiers: random forest, logistic regression, and SVM.
3) When working on a target building having BAS points to be classified, the local clustering structure is exploited for each BAS point based on the point name features represented by k-mers. In this step, Hong et al. [32] use the infinite Gaussian mixture model [35] to eliminate the need to predefine the number of clusters.
4) The same data features as described in Step 1 for the source building are determined for the time-series data of the target building. These data features from the target building are used as the inputs to the trained classifiers from Step 2 to obtain the point types in the target building. A single ensemble prediction of point class is obtained as the weighted arithmetic mean of the predictions from three individual classifiers. The weights assigned to the classifiers are computed for each BAS point according to the similarity between the classifier's prediction and the BAS point's local clustering structure from Step 3.

Table 3
The performance of the point type classification method of Hong et al. [32] using transfer learning between different source buildings and target buildings. NA: not applicable; NR: not reported.

Source buildings	Target buildings								
	A			B			C		
	Accuracy	F1	Coverage	Accuracy	F1	Coverage	Accuracy	F1	Coverage
A	NA	NA	NA	0.943/0.934	0.936/0.931	0.45/NR	0.977/0.970	0.981/0.971	0.76/NR
B	0.807/0.875	0.932/0.913	0.26/NR	NA	NA	NA	0.950/0.952	0.939/0.937	0.49/NR
C	0.862/0.862	0.864/0.864	0.37/NR	0.726/0.702	0.691/0.726	0.28/NR	NA	NA	NA
A + B	NA	NA	NA	NA	NA	NA	098	NR	0.89
A + C	NA	NA	NA	0.899	NR	0.458	NA	NA	NA
B + C	0.863	NR	0.397	NA	NA	NA	NA	NA	NA

The approach was evaluated with a data set consisting of about 2500 points from three commercial buildings located on two different university campuses. The BASs of the three buildings come from different BAS vendors. Twenty-one BAS point types (Table 2) are used for classification. For all 6 possible inter-building tests, the overall prediction accuracy increased and coverage decreased with increasing values of the threshold for the average similarity score. The performance for the ensemble approach in transfer learning between buildings as measured by both accuracy and F1 score exceeded the performance of the random forest classifier, the best baseline (single-method) classifier, in nearly all instances reported. Based on the empirical results, the authors recommend a threshold value of 0.4 to strike a reasonable balance between accuracy (70%–90%) and coverage (30%–70%).

The sensor type categories used in Hong et al. [28,31] are compared in Table 2, and the testing results are summarized in Table 3. In Table 3, for cells with two numbers, the right one indicates the metric value for the simplified approach using only the random forest classifier and time-series data features. The results disclosed two issues. First, for the BAS points that can be classified, although the approach consistently showed increases in both performance metrics (accuracy and F1 score), the increases compared to the values of the metrics for the simplified approach features were very small (e.g. the largest differences were 2.5% and 2.1% for the accuracy and F1 score, respectively). Second, the approach described in Hong et al. [32] showed only small marginal improvements in accuracy and coverage (maxima of 2.9% and 10%, respectively) across the three buildings when the training data were expanded from one building to two.

In another study, Balaji et al. [34] combine the use of point textual information and time-series data to classify point types using a combination of hierarchical clustering and random forest classifiers in a framework known as Zodiac. The textual information, including the point name, description, unit, and data type, are used for hierarchical agglomerative clustering based on the *bag of words model*[3] and inherent patterns in sensor metadata to guide the selection of training examples, which are provided by a domain expert. The time-series data are used to extract four classes of features based on scale, pattern, texture, and shape. These features form the inputs to a random forest classifier, which labels points by type. Zodiac was applied to four buildings on a university campus together having 20,000 points. Most of the equipment in the campus buildings is installed by a single vendor. Results show an average accuracy across buildings of 98% with a decrease of 28% in the number of examples required compared to the regular expression (regex) based method, which requires excessive domain expertise. By comparing results for learning methods based on point metadata in text alone, time series data alone, and both data types, the authors show that 1) use of text metadata alone is substantially more accurate than using time-series data alone and 2) the addition of time series data significantly boosts the accuracy of the method based on only text metadata.

4.1.4. Point type classification based on other approaches

Zhou [24] used pattern recognition to distinguish among sensors. The approach is based on the assumption that a specified sequence of control actions (e.g., opening and closing of valves and dampers and on/off commands for fans and pumps) should result in predictable, identifiable, and unique patterns of changes of physical properties (e.g., temperature, humidity, and flow rate) at specific system locations as measured by sensors. Zhou used test sequences of AHU operation (Table 4) to generate temperature time-series data, which were categorized using a structural pattern recognition technique based on sequences of pattern primitives to recognize the sensor types. After running the six steps in Table 4 sequentially, a structural pattern algorithm is used to recognize the measured temperature patterns for each AHU sensor. A time-ordered series of pattern primitives representing the measured temperature pattern is then matched with template pattern tables developed from experimental testing to distinguish sensor types. Special rules based on common knowledge and observations of testing data are also used to recognize sensor types. For example, when an AHU switches from mechanical cooling off to on, all temperature sensors change except the outdoor-air temperature sensor, thereby leading to the conclusion that the temperature sensor with a constant value corresponds to the outdoor-air temperature. The approach is promising for distinguishing among sensors for a specific system or equipment, but it only addresses part of the automated point mapping problem. The other part, the problem of detecting associations of sensors and actuators to a specific system or equipment, is not trivial.

Similarly, Furst et al. [36] implemented a human-in-the-loop and crowd sourced approach for labeling BAS points that can be directly affected by occupant actions. Underlying this approach, occupants need to provide physical and digital inputs either by control actuations (e.g., turning on or off light switches) or by reading sensors (e.g., the room temperature of a thermostat). Each occupant-provided value is then compared to the BAS point values, and the ones of the same BACnet type and value represent potential matches. The process is iteratively repeated. Each subsequent time, a user inputs another value for the same point until only one matching point remains. For example, if an occupant switches the lights in a space on or off, the BAS points corresponding to manual light switches that experienced the same change of lighting status at the same time as the occupant reported action represent potential matching points. The BAS point (after adequate iteration) is then matched to the light switch activated by the user. The BAS point can also be labeled further as corresponding to other metadata that is associated with the specific light switch activated by the user (e.g., the number of the room in which the switch is located). Furst

[3] The bag of words model in natural language processing is a model in which text is represented by the multiset (bag) of the words in it without regard to grammar or word order. Elements (e.g., words) in the bag can appear multiple times. The number of occurrences of each word in a sample of text (e.g., a sentence or document) is a commonly used feature in bag of words. Those word counts can be normalized and used to compare documents (or sentences) and assess their similarities.

Table 4
Six testing steps for AHU temperature sensor recognition used by Zhou [24].

AHU component	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Supply fan (on/off)	On	On	On	On	On	Off
Return fan (on/off)	On	On	On	On	On	Off
Outdoor damper (% open)	0%	0%	0%	0%	100%	100%
Return air damper (% open)	100%	100%	100%	100%	0%	0%
Exhaust air damper (% open)	0%	0%	0%	0%	100%	100%
Heating water pump (on/off)	Off	Off	On	On	Off	Off
Heating water valve (% open)	0%	0%	100%	0%	0%	0%
Chiller water pump (on/off)	On	On	On	On	On	On
Chiller water valve (% open)	0%	100%	100%	0%	0%	0%
Supply fan speed (%)	100%	100%	100%	100%	100%	100%
Return fan speed (%)	100%	100%	100%	100%	100%	100%

Note: Highlighted cells identify changes from the previous step.

et al. [36] evaluated the approach on a 13,100 m^2 (141,000 ft^2) university campus building. Results presented for a light switch showed convergence to a single point in one iteration in a minute and two and three iterations over 1 to 1½ days for temperature sensors. Though the approach is straightforward, applying it in the field is likely a challenge because 1) it needs intensive participation by occupants and 2) many building points (e.g., static pressures in ductwork) cannot be directly influenced or observed by occupants.

4.2. Human-in-the-loop metadata transformation

Bhattacharya et al. [37–39] transformed BAS sensor point names to metadata in a common namespace by applying the technique of automatically learning the formulation of inductive inference rules from expert-provided examples of point name parsing. Based on the parsing of a relatively small number of example points by a human expert, transformation rules can be learned and applied to other points in the same building and across buildings. They used a combination of clustering and domain-specific language constructs to achieve the necessary robustness to parse building sensor metadata from the heterogeneous and noisy data sets of buildings.

Fig. 4 shows the process of transforming point names (i.e., tags) into sets of individual metadata, which are mapped into a normalized common namespace [27]. First, raw building sensor metadata are syntactically clustered into groups with the members of each group likely sharing the same data schema. An example point is selected from a cluster and provided to an expert to parse into the specified common data namespace. Rules are then synthesized based on the parse

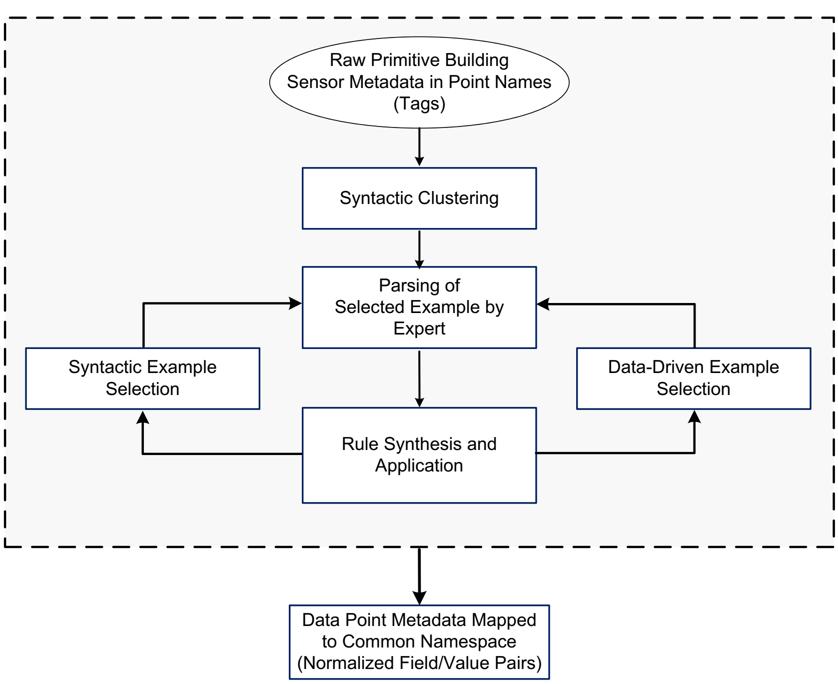

Fig. 4. Primary sub-processes and the relationships between them are shown for Bhattacharya et al. [27].

provided by the expert using a domain-specific language, which can account for many possible encodings of the metadata in sensor names. The resulting rules are used to parse the remaining sensor tags in the cluster from which the example was selected. The process is repeated until a sufficiently large fraction of the metadata for the data points is correctly parsed.

When data streams are available for the points associated with metadata tags, learning can be used for data-driven example selection (the feedback loop on the right-hand side of Fig. 4) to find semantically-related points that have different tags, which the authors refer to as "boosting." The process builds a random forest classifier from the set of sensors that were already classified and applies it to the sensors that have not been classified yet. The authors focus on the use case of obtaining data from a BAS for a third-party external software application. Ordinarily, in this case, all sensors in the building do not need to be classified, only those required by a specific software application, thus decreasing the problem scope somewhat.

The algorithm was tested with data from three commercial buildings, which are located at 3 different institutions and have different building systems, systems installers, and BAS vendors. The three buildings have 1586, 2522, and 1865 sensor points.

The ground truth for all sensor tags was established manually, and the expert was simulated by providing the ground truth parse when the expert's parsing of a BAS point was required. A sensor tag is fully qualified when (1) the correct fields and field values are extracted, (2) no extra incorrect field is identified, and (3) the field-values explain every alphanumeric character in the sensor tag. The effect of each sub-process was tested and evaluated; descriptions of the findings follow.

Pre-processing the data by clustering provides distinct advantages over not using clustering. With clustering, all sensors in the data set were fully qualified after sufficient interaction with the expert. In a plot of fraction of sensors fully qualified versus the number of expert examples, the fraction increases continuously towards 100%, first very rapidly and more slowly as the number of expert examples increases. Without clustering, the fraction of sensors fully qualified increases rapidly, reaches a peak, and then drops precipitously to a relatively constant value that changes at a very slow rate as the number of examples continues to increase. For example, without clustering for Building 3, the fraction fully qualified rose rapidly to over 70% at about 30 examples, then decreased to nearly 20% by approximately 35 examples, increased gradually to a little < 30% by about 130 examples, and remained constant through the 220 examples examined. In contrast, with clustering the fraction increased to about 80% at 50 samples, ~95% at 100, and 100% near 200. For Building 1, without clustering, a decrease in the fraction of sensors fully qualified decreased at about 100 examples and settled at 90% fully qualified. For Building 2, the parsing without clustering was able to fully qualify all sensors, which the authors attributed to much less noisy sensor schemas for Building 1.

Results for the fraction of sensor tags in which each field appears showed that across all three buildings about 20 fields appeared in many sensor tags and many fields occurred in only a very small number of sensor tags (i.e., obscure fields). The authors hypothesized that without clustering, erroneous fields were applied to previously fully qualified sensor tags causing the precipitous drop in the fraction of sensor tags fully qualified. With clustering, this was prevented by rules only being applied to the clusters for which they were developed. Without clustering, application of the rules to sensor tags in clusters other than those for which they were developed (over-generalization of the rules) led to erroneous fields being misapplied because the rules are applied to all sensor tags.

Bhattacharya et al. [27] considered four *methods of example selection*, selection of an example randomly (random), selection of the example with the minimum tag length left to qualify (minleft), selection of the example with the maximum tag length left (maxleft), and selection of a tag with the most frequent unqualified substring (sameleft). The rates of qualification versus number of examples for all methods generally followed the same trajectories with little effect on the number of examples required for full qualification.

Bhattacharya and colleagues also examined *application-oriented qualification* by use of syntactic example selection and by data-driven sample selection. Considering a rouge zone application, which detects zones having air temperatures constantly above the required set point so that cooling is constantly required, sensors with the field "zone temp sensor" or "zone temp setpoint" are needed. Using syntactic example selection alone for Building 1, 147 expert examples were required to fully qualify all the required sensors. Data-driven example selection requires a training set with both positive and negative instances, which are provided by application of syntactic example selection first. Thereafter, the data driven classifier can be used. The data driven approach applied to the same building required only 24 expert examples to fully qualify all needed sensors, a reasonable number of expert examples for practical application and a decrease by a factor of 1/6 in the number of examples required. Similar numbers of expert examples were also found for applications for finding stuck dampers and locating inefficient air-handling units (AHUs).

The results from the three applications (i.e., finding rogue zones, stuck dampers, and inefficient AHUs) on two buildings are shown in Fig. 5. Based on the parsed BAS point names, the applications identified that 1) about 2.5% (5 out of 201 in Building 1 and 2 out of 78 in Building 2) of the total thermal zones are rogue zones; 2) 2.9% of zone dampers are stuck in Building 1 while no damper is stuck in Building 2; and 3) half of the AHUs in Building 1 are deemed inefficient (an AHU is considered inefficient if it serves both rogue zones and overcooled zones) while the application cannot run on Building 2, which does not have the relationship between zones and their corresponding AHUs encoded in point names. These results are examples for deployment of external software applications based on BAS point name parsing. It is not clear whether the application results are the same as the ground-truth number for rogue zones, stuck dampers, and inefficient AHUs, because the ground truth was not reported by Bhattacharya et al. [27].

4.3. Discovery of spatial relationships

In general, point type classification does not extract contextual information. Therefore, BAS point type classification (as described in the studies reviewed in Section 4.1) alone does not answer questions related to the spatial relationships between sensors and building spaces. However, certain software applications built on BASs may need to identify sensors that are associated with the same building space (e.g., HVAC zone or room), leading to the necessity of mapping sensors to building spaces and equipment serving the spaces. In this regard, many studies derive the spatial relationships from different perspectives. The work by Bhattacharya et al. [21,27], discussed in Section 4.2, relied on an automated metadata construction technique to match fields and field values. For example, if sensor names use RM to represent the field

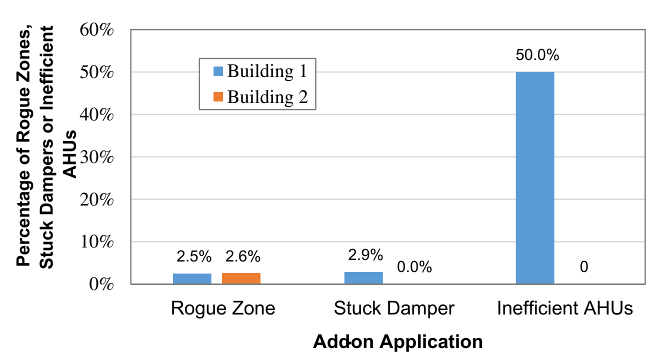

Fig. 5. Results from the deployment of three add-on applications to two buildings by Bhattacharya et al. [27].

“room” followed by a number to represent the field value (i.e., room number), all sensors with the same room number in their names are regarded as associated with the same space. This approach assumes that spatial information is embedded in the point name or associated metadata, which may not be valid in many situations. This limitation could be addressed by using sensor measurements instead of or in addition to sensor names to identify spatial relationships.

Taking this approach, Fontugne et al. [40] used power meter data from lighting and heat pumps to determine the spatial relationships between the associated devices (i.e., lighting fixtures and heat pumps). Using data from nearly 700 sensors of various types (temperature, pressure, device electric power consumption, etc.) in a 12-story building on a university campus over a 3-week period in summer, they find that correlation analysis based on the raw time-series sensor measurements does not identify meaningful relationships among sensors. They then apply the Empirical Mode Decomposition (EMD) technique to decompose the original time-series data into a set of intrinsic mode functions (IMFs) and run correlation analyses on IMFs for pairs of sensor time series. An IMF is a function that has the same number of zero crossings as extrema and a symmetric envelope. EMD enables removal of low frequency trends, which add noise, and focusing on high frequency behavior that is spatially correlated. The results from Fontugne et al.'s analysis show a direct relationship between IMF correlation and spatial proximity: devices close to each other have higher correlation coefficients than devices further apart. A limitation is that building data sets, like those examined in this study, must extend over several days because the underlying trend is daily (a characteristic of the building domain).

Hong et al. [41] also explored the EMD approach for identifying the relative spatial locations of sensors (i.e., which are in each room). Their empirical study on sensor data collected data from 15 sensors across 5 rooms on 4 different floors of a large office building. Each room has three sensors: a temperature sensor, a CO_2 concentration sensor, and a humidity sensor. They found that for all five rooms, the threshold value of the IMF correlation coefficients that can be used to correctly determine 80% of relationships between sensors in different rooms fell into a narrow range between 0.10 and 0.12. The threshold value can be used for all rooms regardless of the sensor type provided there are at least two weeks of data. A clustering algorithm based on IMF correlation coefficients achieved 93.3% accuracy in determining which specific sensors were in each room cluster (although additional metadata are necessary to identify the particular room, e.g., by room number, which the investigators had).

In addition to the studies [40,41] on spatial relationships between sensors or devices and building spaces, Koc et al. [42] performed a preliminary study on inferring the spatial relationships between sensors. They investigated whether linear correlation or statistical dependence is better suited to infer spatial relationships between discharge air temperature sensors and the zone temperature sensor in different rooms of a building. They tested use of the linear correlation measured by the Pearson Product-Moment Correlation Coefficient (PCC) and the statistical dependency measured by the Distance Correlation Coefficient (DCC). Although they concluded that the PCC linear correlation performed better than statistical dependence based on the DCC, the accuracy of the PCC analysis varied significantly from 40% to 100% with temperature sensor pairs and data set sizes.

In the context of smart home applications, Lu et al. [43] developed a tool called Smart Blueprints to automatically map out the floor plan of a house and device locations on the floor plan by analyzing data from smart home sensors. The driver for this investigation was developing a methodology that would enable automatic configuration of home energy systems, specifically to support do-it-yourself installation of sensors with little attention to precise location and no manual configuration necessary. The study specifically focused on installation of sensors for a home energy management system to control lighting and heating. The Smart Blueprints process involves three steps: 1) analyzing sensor data to determine the number of rooms in a home and the devices in each room, 2) inferring room adjacency, which constrains the relative location of rooms, and 3) identifying the floor plan that best explains the sensor data. Sensors are paired to provide additional constraints for the floor plan and the sensor layout, when the sensor data do not provide sufficient constraints. For example, magnetometers and light sensors on windows are used to infer the orientation of doorways, the orientation of windows, the presence of windows in the rooms, and the room orientation. They pair light sensors and motion sensors so that use of light can be identified with a particular room based on activity detected by the corresponding motion sensor. The paper presents the development of inferencing processes for the number of rooms in the home and room adjacency, structural constraints for using magnetometers to infer doorway orientation, using light sensors to infer the orientation of windows, and using light sensors on door trim to infer the presence of windows in rooms.

Searching for the floor plan requires defining the search space and reducing the search space size. The search spaces before applying constraints were shown to be as large as the order of 10^{18} initially for the test homes. The authors describe several methods used to prune the search space, which decrease it to 10^6 to 10^9 possibilities. The last pruning technique is α–β pruning applied to a score assigned to each node based on a non-monotonically increasing function. Low scores are preferred and subtrees with roots having higher scores than an identified leaf node with a low score are eliminated from further consideration. Two structural heuristics are applied, one to “minimize the number of door and neighboring rooms that share a wall with a window” and the other to minimize the number of corners of a building. The score of each floor plan is increased by one for each corner it has greater than four. Application of α–β pruning and the heuristics reduces the search space to between tens and thousands of potential floor plans. Then only plans with low scores are considered further. Most of the remaining floor plans are eliminated by filtering based on internal inconsistencies. The floor plans remaining after filtering are shown to the homeowner, who can select the correct floor plan from these candidates.

The method was tested on seven occupied houses and was able to find a small number of candidate floors plans (between 1 and 5 for most test cases) based on 1 to 2 weeks of sensed data. Four scenarios of different structural constraints and sensor locations/orientations were explored to determine their effects on the number of final floor plan candidates. To increase the size of the testing sample, 22 additional floor plans were obtained from the web and used to evaluate the search space filtering. Ten or more candidate floor plans were obtained for only three of these cases, the average number of candidates was three, and all of the candidate groups included the actual floor plan. A major disadvantage of the Smart Blueprints they developed lies in its stringent requirement on sensor types, density, and orientation of placement. For example, multiple lighting sensors are used in each room, and some sensors need to be placed behind the door jamb of a doorway and on windows (although the research team developed solutions for these installations). The methodology had trouble with certain configurations. For example, the method at the time of reporting was unable to handle multiple-story homes, and open floor plans presented difficulty. The method as presented also did not support homes with unaligned walls, angled doorways, and rooms not joining at 90 degrees. Home floor plans with insufficient constraints present especially difficult cases. Despite these limitations, the accomplishments presented would likely serve significant fractions of home floor plans.

In contrast, Ellis et al. [44] developed a method to automatically determine room connectivity with few requirements on sensor density and placement. The method requires only one lighting sensor, which measured at a 5-second sampling interval, and one motion sensor, which measured at a 1-second sampling interval, for each room. The two sensors were part of a single sensor node (module), which also included sensors for temperature and humidity. The principles for room

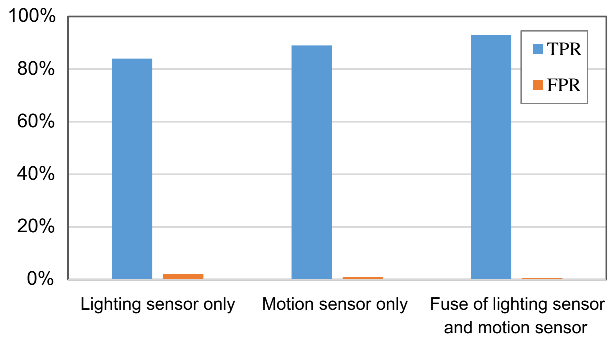

Fig. 6. True positive and false positive rates (TPR and FPR, respectively) found in testing by Ellis et al. [44] of a method to automatically determine room connectivity using two data streams individually and fused together.

connectivity inference are based on the assumption that when room lighting is switched on or off, the lighting levels in connected rooms also change as the result of light "spillover." Thus, rooms with simultaneous light level changes indicate that they are connected. Furthermore, as occupants move from one room to another, the motion sensor in the first room ceases to detect movement while the sensor in the second room starts to detect it. Thus, a motion event ending in one room followed by another motion event starting in another room indicates that those two rooms are connected. Ellis et al. [44] tested their method using lighting only, motion only, and fusing of the two data streams in both homes. Results for use of the individual sensors were an 84% true positive rate and a 2% false positive rate for use of lighting sensors only and an 89% true positive rate and a 1% false positive rate for use of motions sensors alone. Use of both data streams resulted in improvements to 93% true positive rate and 0.5% false positive rate for all room connections (see Fig. 6). One week of data was sufficient to achieve this accuracy.

4.4. Discovery of functional relationships

Our review of literature in the field of semantic discovery and mapping of control system points found only one study [45] that investigated the problem of automatically discovering functional relationships between HVAC equipment. The study explored methods for discovering the functional (connected to or served by) mapping of air handling units (AHUs) and the variable-air-volume (VAV) terminal boxes to which each AHU provides cooled air in a building heating, ventilating, and air-conditioning (HVAC) system. The methods are all applied to a commercial building having 3 chillers, 4 AHUs, and 179 VAV boxes serving the thermal zones.

The investigators compared four methods:

1) correlations of raw data for zone-air temperature, air damper position, and VAV reheat valve position,
2) correlation of two principal components determined from feature vectors with the VAV supply-air temperature, where the feature vectors included the zone-air temperature, the air flow-rate set point, the day of the week, and the time of day,
3) a novel approach based on system identification (from the control field),
4) a new method based on observing VAV box responses to perturbations of the supply-air temperature set point of each AHU.

The system identification method (item 3) used a time-series regression model for the temperature of the zone served by VAV terminal box i at time t ($T_{zone,i,t}$) based on the physics of the heat transfer in VAV systems, namely,

Table 5
Fraction of VAV boxes correctly classified; Pritoni et al. [45].

Method	Fraction of VAV boxes correctly classified (%)
Correlations of raw data	38%
Correlations of principal components	32%
System identification approach	32%
New perturbation method	79%

$$T_{zone,t} = \beta_1 T_{zone,i,t-1} + FLW_t \sum_{t}^{t-k} \beta_{2i} RVP_i + \beta_3 FLW_t T_{sa,t}^{AHU} + \beta_4 T_{oat,t}, \quad (1)$$

where FLW_t represents the AHU flow rate, RVP_i is the reheat valve position for VAV box i, $T_{sa,t}^{AHU}$ is the temperature of the supply air from the AHU at time t, $T_{oat,t}$ is the outdoor air temperature at time t, the subscript i designates terms that correspond to the ith VAV terminal box that serves the zone, the summation in the second term on the right hand side is over all times steps from k before time t through time t, and β_1, β_2, and β_3 are the coefficients determined by fitting the model to the HVAC system data for which both linear regression and Lasso [46] were used on 15-minute sampled data.

The results as the fractions of correctly classified VAV boxes are shown in Table 5 for the four methods; the most successful of the first three methods was correlation of the raw data with a correct classification rate of 38%.

Pritoni et al. [45] also developed and tested a new method (the fourth in Table 5) based on perturbing the supply-air temperature of each AHU separately and detecting the VAV boxes connected to it based on distinguishable signals from them without negatively affecting the comfort of occupants of the connected zones. Because the characteristic thermal times of HVAC systems are long, each operating mode was held fixed for a full day, and the AHU supply-air temperature set point was held at 52 °F (11.1 °C) for the cold mode and at 60 °F (15.6 °C) for the hot mode, respectively only 5 °F (2.8 °C) below and 3 °F (1.7 °C) above the 57 °F (13.9 °C) ordinary supply-air set point for the building in which the method was tested. The cold and hot modes were implemented alternately on sequential days, and data were resampled daily (i.e., average daily values of measured variables were used). Average daily vectors are formed from measurements of RVP, T_{zone}, and damper position and standardized (mean = 0 and standard deviation = 1). The Euclidean distance between hot and cold vectors for each pair of hot/cold perturbations is used as a metric for each VAV-AHU combination. For each VAV box, a "vote" (rather than independent classification) is used to select the AHU with the largest metric, which is expected to be the AHU associated with the specific VAV box.

The method identified the correct AHU for 79% of the VAV boxes, double or better the success rate for the other methods examined (Table 5). The authors reasoned that three primary situations led to incorrect classification of AHUs to VAV boxes: 1) VAV boxes with little or no response to the perturbations, e.g., potentially caused by a poorly tuned VAV-box control loop, 2) VAV boxes that had unexpected physical behavior, e.g., associated with large unexpected disturbances, and 3) boxes that responded most strongly to an incorrect AHU perturbation.

Although the results for the new method are very encouraging, applications of the method to a larger set of buildings with single-duct VAV HVAC distribution systems and to relationships among other HVAC and building components are needed.

4.5. Concluding remarks on the use of machine learning

In all the previous work reviewed that used machine learning, very simple models are used (e.g. latent semantic indexing (LSI) [47] or Gaussian mixture models (GMM) [48]), but the work still manages to achieve good results on their limited data sets. LSI is a model for determining the different topics in a corpora of text documents, but using it on a few tens of characters (point names) is rather awkward. Newer models, such as focused topic models [49] and character recurrent neural networks (RNN) [50] would be more appropriate. GMMs are often used for clustering real-valued data (not discrete-valued, e.g. text). It is less common to use them for classification, since many good models/algorithms have been built specifically for classification (e.g., support vector machine [51]). Using more appropriate models will improve the results. Developing new models that capture more of the problem space will also improve the results (e.g., simultaneously modeling point names and connections). Recently, machine learning researchers have shown that it is possible to predict an image from a caption, or a caption for an image [52]. The combined text + image model also performs better in classification tasks when classifying images/text both separately and together. Comparing point-mapping to text + image modeling is appropriate since the point name is a short (potentially obfuscated) description of the point (just as a caption is a short description of an image) and point time-series data are commonly processed with the same methods as images (e.g., using Wiener or wavelet filters) where images and time-series are treated as signals.

In addition to high accuracy and coverage, an automated point mapping solution must be scalable to significant fractions of the building population to keep the number of individual algorithms needed to cover most of the population of commercial buildings having BASs manageable (see Section 3). The research reviewed in this paper has done some preliminary testing mostly on relatively small data sets of limited diversity (e.g., data from buildings on one or two university campuses). It remains to be shown that the methods developed are scalable, but the approach taken of initially testing on small data samples before testing at scale is common practice in machine learning. However, the question of scalability may be premature with regard to automated point mapping because most of the methods developed in the work reviewed focused on only a single sub-process that might be part of automated point mapping rather than the entire process. Regarding the relationship of scalability to the size and complexity of the data set used for initial testing, we consider image recognition. It is typical in image recognition to perform basic tests (e.g. to tune model hyper-parameters) on a rather small data set (e.g., 60 thousand 28 × 28 images), then to generalize the findings by testing on larger data sets (e.g., 2 million 256 × 256 images). Therefore, after adequate initial testing on small data sets like those used in the research reviewed, testing will need to transition to larger, more complex data sets.

5. Key technical needs for realizing automated point mapping

Development of a methodology for automated control system point mapping that meets the requirements identified in Section 3 of this paper—agnostic to BAS and third party software applications, support all semantic information needed by third party software and hardware and BASs themselves, accuracy for determination of point semantics near 100%, coverage of a large fraction of all BAS points and metadata associated with them, minimal human involvement, and scalability to most of the commercial buildings sector —represents the primary technical challenge to automated BAS point mapping. Three key groups of technical needs with related research questions critical to addressing this primary challenge are described in this section: 1) complete solution based on meticulous problem formulation, 2) robust, verified approaches, 3) test cases and performance metrics and open data sets for a diverse set of representative buildings to support consistent performance testing and reporting, and 4) an understanding of the applicable data space and ensuring future adaptability of automated BAS point mapping.

5.1. Need for a complete solution and underlying problem formulation

Bhattacharya et al. [21] analyzed 87 third-party software applications and identified a variety of semantic information needed by those applications. The semantic information that is usually needed by third-party software built on BASs includes, but is not limited to, the following: point type (e.g., temperature sensor, thermostat set point, and fan speed command), specific equipment (e.g., AHU 5 and VAV terminal box 23) with which points are associated, functional relationships between equipment (e.g., an AHU and its connected VAV terminal boxes), and spatial relationships among points, equipment, and building spaces (e.g., an air temperature sensor is located in room 101, which is served by VAV terminal box 23). Therefore, ideally, automated point mapping should be capable of extracting all semantic information needed by third-party software. However, based on our literature review, previous studies investigated different elements of the automated point mapping problem (e.g., sensor type classification, sensor location identification, and discovering functional relationships). Although all these efforts are important steps towards a complete, practical solution to automated point mapping, a comprehensive solution for building control systems that is adequately scalable, accurate, applicable across applications, and sufficiently tested does not yet exist. Some important questions to be answered include:

- *What is the optimal balance of competing automated point mapping characteristics?* The characteristics include, for example, the degree of automation, the accuracy of the method, and its coverage. A target of entirely automating point mapping without humans in the loop might seem reasonable. However, the optimal degree of automation depends on the cost and difficulty of full automation compared to using humans either to provide some of the information (e.g., training data) to enable an automated process or to supplement incomplete automated mapping of metadata (e.g., 90%) with the remaining metadata (e.g., the last 10%) mapped manually. The accuracy of an automated process represents another critical factor in evaluating solutions. A method that maps 100% of the necessary information but with accuracy well below 100% could leave many metadata (and points with which they are associated) incorrectly mapped, leading to errors in control or other processes that depend on the data. The quantified performance of point mapping (e.g., the percentage of points with accurately defined semantics) and its impacts must be considered in developing a solution. Ultimately, the user community will determine market-compatible answers to how much automation is optimal, what fraction of the metadata needs to be covered, how accurate the automated process needs to be, and other quantified "optimal" characteristics.
- *What metadata needs to be addressed by automated point mapping?* Both the breadth and depth of metadata need to be considered. Here, breadth refers to the completeness of metadata for BAS points, and the depth refers to the resolution at which metadata is provided. For example, sensor type and sensor location represent different kinds of metadata for a sensor in the breadth dimension. Greater breadth is indicated by more types of metadata. The depth of metadata can be illustrated by comparing the resolution to which the location of a sensor is specified in two names for the same sensor point, "Building 1 Temperature Sensor" and "Building 1 Room 1245 East Wall Air Temperature Sensor"; the former only indicates that the temperature sensor is located in Building 1, while the latter specifies the location to a greater resolution—east wall of room 1245 in Building 1. In general, the difficulty of point mapping increases with increasing breadth and depth of the required metadata. A scalable automated point mapping solution must cover sufficient breadth and depth to support mapping of data across broad categories of buildings, system types, geographic locations, sensors, actuators, control points, and types of software applications, while requiring little labor or special expertise to keep the costs of

installation and use low. Furthermore, to simplify and promote wide use, the "discovered" metadata need to be stored in one or more open (and ideally standardized) data schemas. Presently, significant gaps exist between available metadata schema and the needs for BAS point sematic information representation [21].

5.2. Need to align methods with the actual point mapping problem

Many approaches have been proposed to address the problem of BAS point mapping, and encouraging results on the algorithm performance have been reported (see Section 4). The following questions need to be considered in developing algorithms that align with actual systems and conditions in commercial buildings:

- *What sources can be used for metadata?* Most previous studies have used point names and/or point time-series data to extract semantics for BAS points. Balaji et al. [34] and Parks [30] extended metadata sources to include units, data types, and text descriptions. Other sources of metadata (e.g., log messages and schedules) and results from physical testing, such as demonstrated by Pritoni et al. [45], represent additional potential sources of metadata. Judicious use of system perturbation may be essential for complete automation of building control system point assignment and system configuration (e.g., starting with the physical system but few, if any, named points).
- *What alternatives exist to a priori definition of data features?* In previous studies, textual data (e.g., point names) have been represented by bag of words models [29,34] and k-mers [32] while time-series data features have been represented by statistical features in a more diversified manner (e.g., [32,33]). Alternative methods defining data features should be explored; for example, developments in machine learning approaches such as Latent Dirichlet Allocation [53], focused topic modeling (FTM) [49,54], and recurrent or convolutional neural networks (RNN, CNN) [55,56] should be considered for learning features from data rather than predefining them.
- *What deficiencies in BAS data do automated point mapping methods need to accommodate and how?* Any successful automated point mapping solution must have the ability to use BAS data as it exists in actual systems. All previous studies have recognized problems associated with point names and other attributes (e.g., units and object types), including multiple and inconsistent uses of schemas, abbreviations, and delimiters across buildings and even among points for the same building. Buildings also suffer operational faults and hence the time-series data may be noisy and even represent faulty behavior. Little work has been done to understand the effects of data from faulty physical systems on point mapping algorithms and how to ensure that algorithms are robust to such data. Consideration of faults also leads to the question of how data collected over a time period during which physical system faults are repaired will affect algorithm performance. Furthermore, whether and how data sampling time intervals affect the performance of point mapping methods has received little attention.
- *How well do the principles underlying approaches match the mapping problem?* Although there are increasing numbers of studies on automated point mapping, most approaches to date rely on clustering and discriminative classification algorithms. Often common existing machine learning methods are directly used or adapted with different parameters or different combinations of algorithms. These methods group points into classes (e.g., sensor type), which is important information but not sufficient to uniquely define each point as required for point mapping. Specialization of machine learning algorithms may be needed to automate point mapping. Furthermore, assumptions underlying each method need to be thoroughly vetted for applicability to actual building systems and documented. For example, the assumption that sensors located spatially close to each other show stronger correlations than those further apart is not valid for many building control system points. Instead functional relationships between points often have stronger correlations than spatial proximity.
- *What methods are appropriate for point mapping for setting up new BASs?* As mentioned in Section 2.1, point mapping is needed in setting up new BASs to ensure that sensors and actuators are correctly wired to controllers and their data points are bound to BAS graphics, and to streamline the naming of points consistently by automating this process. Because no human readable point names exist initially in a new BAS, the problem of point mapping in new BASs is different and more difficult than the problem of connecting external third-party software to an existing BAS. A concept to address this situation might involve exciting actuation points in systems and equipment and observing the responses of other points [similar to the approach used by Pritoni et al. [45] for identifying the mapping of VAV boxes to air handling units for an existing BAS (see Section 4.4)], after first manually identifying, naming, and attaching metadata to a small number of points. The responses of other points resulting from the excitations could be analyzed to identify unknown points and connections between points. In somewhat related work reviewed in Section 4.1.4 of this paper, Zhou [24] devised a sequence of AHU control actions to generate temperature time-series data, which are categorized using a structural pattern recognition technique to recognize sensor types. However, additional research is needed to address 1) which and how many points need to be manually defined, 2) how to define the excitations to use (e.g., their shapes and durations), 3) what sequence of excitations should be used for different systems and equipment, 4) how the number of excitations necessary depends on the sizes of the physical and control systems, 5) what approaches should be used to analyze system responses, and certainly other issues.

5.3. Need for representative test cases, data sets, explicit test procedures, and consistent performance metrics

Unsurprisingly, all studies examined in this paper test algorithms on relatively small samples of buildings, many sharing common characteristics (e.g., part of university campuses, the same type of HVAC system, etc.) even though some investigators clearly attempt to push these boundaries. However, the results of testing algorithm performance from previous studies are difficult to compare because 1) the specific problems addressed differ (e.g., the degree to which conventions were consistently used in original naming of the BAS points, whether the BASs were from a single vendor and installer or not, the ages of the building systems and control systems, and the metadata targeted) and 2) different metrics are used to characterize performance (e.g., precision, recall, accuracy, coverage, F1 score) and observed behaviors are analyzed differently. The problem of incomparable results from different studies might be addressed by consideration of the following questions:

- *How might an open standard library of test cases and data with adequate diversity of building and BAS characteristics be developed and made widely available?* Representation of the diversity of building types, HVAC and other systems, building and system ages, point naming practices, and other factors is needed to ensure applicability of the methods developed to a significant fraction of the commercial buildings sector with BASs. An open library of test cases that can grow to sufficient size and diversity over time may be critical for convergence to practical, scalable solutions. Such a library could be developed by a cooperating team of organizations and grown, disseminated, and managed using a model analogous to open source software models. Non-technical issues (e.g., privacy concerns associated with revealing the performance or occupancy schedules of specific buildings) may present barriers to obtaining agreement from building owners to include data on their buildings in such a library, but attention to data privacy, for example, using careful anonymization and exclusion of data unnecessary for the intended uses, could mitigate these issues.

- *What performance metrics are essential and should be used to evaluate, report, and compare the performance of point mapping approaches?* As we discussed in Section 4, many previous studies have focused on point type classification with precision and recall used to evaluate classifier performance. In this context, precision[4] indicates the fraction of all classified points that are correctly classified, and recall[5] represents the fraction of all points that should be classified that are classified correctly. In studies that focused on identifying and assigning semantic metadata for control system (e.g., sensor) points, accuracy and coverage were commonly used as performance metrics. Coverage is the fraction of a data set for which an algorithm makes a prediction, and accuracy is the fraction of the covered data in a data set that is correctly predicted. Other metrics can be used to evaluate classifiers/predictors, including sensitivity, specificity, fall out, false discovery rate, false negative rate, F_1 score, G-score, balanced accuracy, correlation coefficients, uncertainty coefficient, informedness, markedness, Cohen's kappa coefficient, the Mann–Whitney *U* test, etc. Each has its benefits and weaknesses. A minimum set of metrics that should be reported in all related work in this field should be agreed upon by the relevant research community and adopted as a standard reporting practice to improve the comparability of results for methods developed.
- *How should the uncertainty in the characterization of the performance of point mapping methodologies be reported?* What minimum accuracy would qualify a method as ready for general field use? The uncertainty in performance of a point mapping method is critical for understanding the impacts of the method on the performance of control systems and third-party software applications that rely on the resultant mapping. In this regard, Hong et al. [32] used the similarity between data feature-based classification and name feature-based classification as an indicator of confidence in predictions and found an intrinsic relationship between confidence and coverage with accuracy increasing and coverage decreasing with increasing values of the threshold for the average similarity score for baseline classifiers. Greater clarity regarding the impacts of uncertainty is needed in general as well as target values for performance metrics (e.g., a value for minimum accuracy).

5.4. Need to understand the applicable data space and ensure future adaptability of automated BAS point mapping

As defined in Section 2.3, one of the steps for connecting third-party software or hardware to an existing BAS needs to establish the relationships between the inputs and outputs of the third-party software/hardware and the corresponding BAS points. Most relevant research in the field (see Section 4) has focused on extracting BAS point semantics from either textual data or time-series data. However, a complete understanding of information requirements by third-party software and devices is valuable in beginning to identify the set of information for which point mapping is needed. Bhattacharya et al. [21] explored the application data space by analyzing a set of 87 applications based on publications in a few venues (the BuildSys, HomeSys, and E-Energy conferences) for the purpose of evaluating three prevalent metadata schema for buildings. They characterized the relationships among metadata by the applications and evaluated whether the relationships could be captured by the three building data schema.

[4] Precision = number of true positive classifications / total number of classifications.

[5] Recall = number of true positive classifications / total number of positive classifications.

Understanding the data space that automated point mapping needs to serve is important, and additional investigation beyond the work of Bhattacharya et al. [21] is needed for this purpose; however, some caution should be exercised regarding the limitations of only defining this space based on existing applications. An inventory of data required by existing applications will define the data needs for those current applications only. New applications will be developed in the future, and automated data mapping methodologies need to be flexible enough to meet their mapping requirements. An alternative to understanding the data space of existing applications could be determining the set of all points used in BASs. Collection of both types of data will likely be important to understand the diversity of data that automated point mapping needs to address. Even this set though is likely to evolve over time; therefore, methods of mapping that can automatically adapt to changes in data mapping requirements with time should be investigated as a means to ensure sufficient flexibility for future applications.

6. Conclusions

Primary conclusions from the review of recent advances in automated mapping of control system points and their metadata documented in this paper are:

- A practical, adequately scalable, automated point mapping method for building control systems does not yet exist.
- Several methods based on text analysis and time-series data analysis have been developed and tested at limited scale with encouraging results. Correct mapping of points has exceeded 70% and in some cases 90%. Coverage, in some cases, approaches 100%.
- Methods developed and tested use primarily point type classification, many for sensors only. One study reviewed used system excitation (i.e., perturbation) to discover a relationship between physical system components.
- Testing completed to date is inadequate to understand the effects of many factors on performance of the methods developed (e.g., the diversity of building types, types of owners, construction type, BAS system vendors, ages, and installers, HVAC system types, factors that vary with geographic location, and the effects of existing equipment and control system faults on method performance, to name some).
- Differences in testing data sets make comparison of results by different investigators difficult to compare.
- A key need in this field is an adequately large and diverse open data set on which existing and new methods can be applied to assess performance of them at scale.
- A minimal set of standard metrics for assessment of performance of all automated methods for metadata discovery/mapping and associated point mapping is needed to increase comparability of testing results. Other metrics could be reported in addition to this minimum set, but the minimum should always be reported.

Acknowledgements

The authors gratefully acknowledge support for the work reported in this paper by the Building Technologies Office of the U.S. Department of Energy.

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